

WEEK 6 – EXPLORATORY DATA ANALYSIS (EDA)

OBJECTIVE:

The objective of this analysis is to explore the Healthcare dataset using Python, Pandas, Matplotlib and Seaborn. The analysis focuses on patient information, medical conditions, billing amounts, insurance providers, admission types, monthly admissions and age groups.

ANALYSIS:

Analysis	Description
Age Analysis	Analysed the age distribution of patients across different medical conditions using box plots.
Gender Analysis	Compared male and female patient distribution across different medical conditions.
Billing Analysis	Compared the average billing amount for different medical conditions.
Insurance Analysis	Analysed billing amount distribution for different insurance providers.
Admission Analysis	Compared billing amounts across different admission types.
Monthly Analysis	Analysed the number of patient admissions for each month.

Correlation Analysis	Studied the relationship between Age, Stay Days and Billing Amount using a correlation heatmap.
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KEY FINDINGS:

- Different age groups were analysed across various medical conditions.
- Gender distribution was compared for different medical conditions.
- Billing amounts were analysed based on medical conditions and insurance providers.
- August recorded the highest monthly admissions with 4,832 patients.
- February recorded the lowest monthly admissions with 4,255 patients.
- Age and Billing Amount showed a very weak correlation of -0.003832.
- Stay Days and Billing Amount showed a very weak correlation of -0.005602.

MAJOR OBSERVATIONS:

- The dataset provides useful information about patients and hospital admissions.
- Billing amounts vary across medical conditions, insurance providers and admission types.
- Monthly patient admissions show some variation throughout the year.
- The correlation analysis shows that Age and Stay Days have almost no linear relationship with Billing Amount.
- Different visualization techniques help to understand the healthcare data more clearly.

RECOMMENDATIONS:

- Healthcare organizations can use EDA to understand patient and admission patterns.
- Billing information can be analysed further to identify factors affecting healthcare costs.
- Monthly admission trends can help in better planning of hospital resources.
- Further analysis can be performed using additional patient and hospital-related variables.

CONCLUSION:

The Week-6 EDA successfully explored the Healthcare dataset using Python, Pandas, Matplotlib and Seaborn. Different charts and statistical techniques were used to analyse patient details, medical conditions, billing amounts and admission patterns. The analysis provides a clear understanding of the dataset and can be useful for further healthcare data analysis.